

NANYANG TECHNOLOGICAL UNIVERSITY

School of Mechanical and Aerospace Engineering

E3.2

VIBRATION OF A TWO-DEGREE OF FREEDOM SYSTEM

**Mechanics of Machines Lab
(N3-B1c-03)**

E3.2

VIBRATION OF A TWO-DEGREE OF FREEDOM SYSTEM

SUMMARY

I) Area of Study

Mechanical vibrations

II) Learning Objectives

By completing this project, students will be able to learn about

a. Theoretical Models, Principles and Concepts

Concept of resonant frequency and mode shape

Free vibration of two degrees-of-freedom system

Forced harmonic vibration of two degrees-of-freedom system

b. Experimental Techniques

Observation of resonant frequency and mode shape of a two-degree-of-freedom system by either impact hammering or adjusting excitation frequency

c. Instrumentation

D. C. power supply

Charge amplifier

Accelerometer

Storage Oscilloscope

Printer

d. Data Analysis

Waveform analysis

Frequency response and graph analysis

E3. 2

VIBRATION OF A TWO-DEGREE OF FREEDOM SYSTEM

1 INTRODUCTION

Vibration is concerned with oscillatory motions of bodies and the forces associated with them. It is initiated when energy is imparted to the mechanical system by an external source.

Vibration occurs in many engineering systems, and if uncontrolled may lead to catastrophic failures. For example, the vibration resulting from the dynamic unbalance of a helicopter rotor blade may lead to the pilot losing control of the helicopter and causing it to crash. Also the vibration of a machine tool will lead to quality degradation of a workpiece.

The number of co-ordinates required to describe completely the positions of all the parts of a system at any instant is called the degree of freedom (DOF) of the system. There are two equations of motion for a two-DOF system, one for each mass. The equations of motion lead to a frequency equation that gives two natural frequencies for the system. If we give suitable initial excitation, the system vibrates at one of these natural frequencies. During free vibration at one of the natural frequencies, the amplitudes of the two-DOF are related in a specific manner and the configuration is called a principal or natural mode of vibration.

If we give an arbitrary initial excitation to the system, the resulting free vibration will be a superposition of the two natural modes of vibration. However, if the system vibrates under the action of an external harmonic force, the resulting forced harmonic vibration takes place at the frequency of the applied force. Under harmonic excitation, resonance occurs (i.e. the amplitudes of the two co-ordinates will be maximum) when the forcing frequency is equal to one of the natural frequencies of the system.

2 OBJECTIVES

- To understand the free vibration of a system having two-DOF
- To understand the forced vibration of the system
- To understand the mode shape of the system

3. THEORY

Figure 1 shows a vibratory system, having two masses, m_1 and m_2 . This is a two-DOF system because two coordinates, x_1 and x_2 , are required to define the motion of the system. The mass m_1 is subjected to a harmonic force $P_0 \sin \omega t$. The equations of motion are:

$$m_1 \ddot{x}_1 = P_0 \sin \omega t - 2k_1(x_1 - x_2) \quad (1)$$

$$m_2 \ddot{x}_2 = 2k_1(x_1 - x_2) - 2k_2 x_2 \quad (2)$$

or in matrix form,

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} 2k_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} P_0 \sin \omega t \\ 0 \end{bmatrix} \quad (3)$$

where x_1 and x_2 are the displacement amplitudes associated with masses m_1 and m_2 respectively.

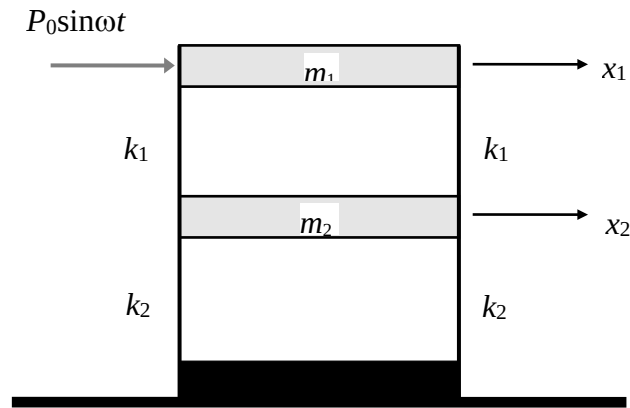


Figure 1 Two-DOF Vibration System.

3.1 DENOTATIONS

m_1 and m_2 are masses of the two-DOF vibration system

k_1 and k_2 are the stiffnesses of the leave springs supporting the masses m_1 and m_2

x_1 and x_2 are the displacements associated with masses m_1 and m_2

$\ddot{x}_1$ and $\ddot{x}_2$ are the accelerations associated with masses m_1 and m_2

A and B are the displacement amplitude associated with masses m_1 and m_2

P_0 is a harmonic force applied to the mass m_1

ω is the angular natural frequency of the free vibration

ω_1 and ω_2 is the first and the second resonant angular frequencies

ϕ is the phase of the vibration

ϕ_1 and ϕ_2 is the phases at the first and the second resonant angular frequencies

A_1 and B_1 are the displacement amplitudes at the first resonant frequency ω_1

A_2 and B_2 are the displacement amplitudes at the first resonant frequency ω_2

r_1 is the displacement amplitude ratio, i.e. A_1/B_1 , at the first resonant frequency ω_1

r_2 is the displacement amplitude ratio, i.e. A_2/B_2 , at the second resonant frequency ω_2

x_{10} and x_{20} are the initial displacements associated with masses m_1 and m_2

v_{10} and v_{20} are the initial velocities associated with masses m_1 and m_2

Ω is the angular speed of a driving motor

m_0 is the unbalance mass of the driving motor with rotating parts

e is a radial eccentricity of the unbalanced mass from the geometric axis of rotation

3.2 FREE VIBRATION

For free vibration, the external force P_0 is set to zero. Equation (3) becomes the homogeneous equations

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} 2k_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (4)$$

We are interested in knowing whether m_1 and m_2 can oscillate harmonically with the same frequency and phase angle but with different amplitudes. Assuming that it is possible to have harmonic motion of m_1 and m_2 at the same frequency ω and the same phase angle ϕ , we take the solutions of Equation (4) as

$$x_1 = A \sin(\omega t + \phi) \quad (5)$$

$$\text{and } x_2 = B \sin(\omega t + \phi) \quad (6)$$

These expressions imply that in a natural mode of vibration both masses follow the same harmonic function, having the angular frequency ω and phase ϕ . A and B represent the steady state amplitudes of masses m_1 and m_2 . Substitution of Equations (5) and (6) into Equation (4) produces the final form,

$$\begin{bmatrix} 2k_1 - \omega^2 m_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) - \omega^2 m_2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (7)$$

One possible solution of Equation (7) is that $A = B = 0$, which corresponds to the equilibrium position and is a trivial solution. The equation can have nonzero solutions only if the determinant of the coefficients of A and B is equal to zero. Thus

$$\begin{vmatrix} 2k_1 - \omega^2 m_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) - \omega^2 m_2 \end{vmatrix} = 0 \quad (8)$$

This determinant gives

$$m_1 m_2 \omega^4 - [2m_1(k_1 + k_2) + 2m_2 k_1] \omega^2 + 4k_1 k_2 = 0 \quad (9)$$

Equation (9) which is quadratic in ω^2 , is called the characteristic equation (also called frequency equation) of the system. It has two roots, i.e.,

$$\omega_1^2 = \frac{[m_1(k_1 + k_2) + m_2 k_1] - \sqrt{[m_1(k_1 + k_2) + m_2 k_1]^2 - 4m_1 m_2 k_1 k_2}}{m_1 m_2} \quad (10)$$

$$\omega_2^2 = \frac{[m_1(k_1 + k_2) + m_2 k_1] + \sqrt{[m_1(k_1 + k_2) + m_2 k_1]^2 - 4m_1 m_2 k_1 k_2}}{m_1 m_2} \quad (11)$$

The two angular frequencies depend only upon the masses and the stiffnesses. These two frequencies are also called as the natural frequencies of the system. Equations (10) and (11) are written in such a manner that $\omega_1 < \omega_2$, where ω_1 is the first natural frequency and ω_2 is the second natural frequency.

Substituting ω_1 into Equation (7), the amplitude ratio r_1 can be calculated

$$r_1 = \frac{A_1}{B_1} = \frac{2k_1}{2k_1 - \omega_1^2 m_1} = \frac{2(k_1 + k_2) - \omega_1^2 m_2}{2k_1} \quad (12)$$

Similarly, when $\omega = \omega_2$, the amplitude ratio r_2 is

$$r_2 = \frac{A_2}{B_2} = \frac{2k_1}{2k_1 - \omega_2^2 m_1} = \frac{2(k_1 + k_2) - \omega_2^2 m_2}{2k_1} \quad (13)$$

These amplitude ratios represent the shapes of the two natural modes of vibration (also called the principal modes).

When the system vibrates at the first natural frequency (ω_1), the amplitudes of masses m_1 and m_2 have to satisfy the amplitude ratio r_1 . Thus, the vibration displacements under the first natural frequency are

$$x_1(\omega_1) = r_1 B_1 \sin(\omega_1 t + \phi_1) \quad (14)$$

$$\text{and } x_2(\omega_1) = B_1 \sin(\omega_1 t + \phi_1) \quad (15)$$

These expressions completely describe the first mode of vibration in terms of frequency and mode shape. As an example, Figure 2 shows the first mode of vibration for $r_1=2$. In this mode of vibration, displacement of mass 1 is always twice as much as the displacement of mass 2. Also masses m_1 and m_2 always vibrate *in phase*.

The second mode of vibration is

$$x_1(\omega_2) = r_2 B_2 \sin(\omega_2 t + \phi_2) \quad (16)$$

$$\text{and } x_2(\omega_2) = B_2 \sin(\omega_2 t + \phi_2) \quad (17)$$

Figure 3 shows an example of second mode of vibration for $r_2 = -1$. In this mode, masses m_1 and m_2 always vibrate *out of phase* to each other with the same amplitude.

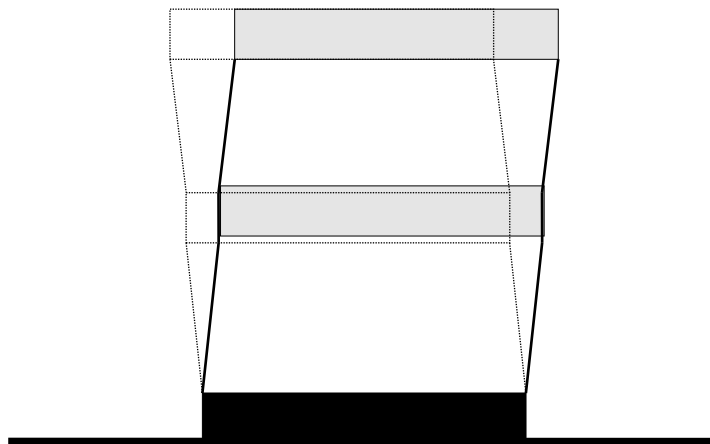


Figure 2 First mode of vibration ($r_1=2$).

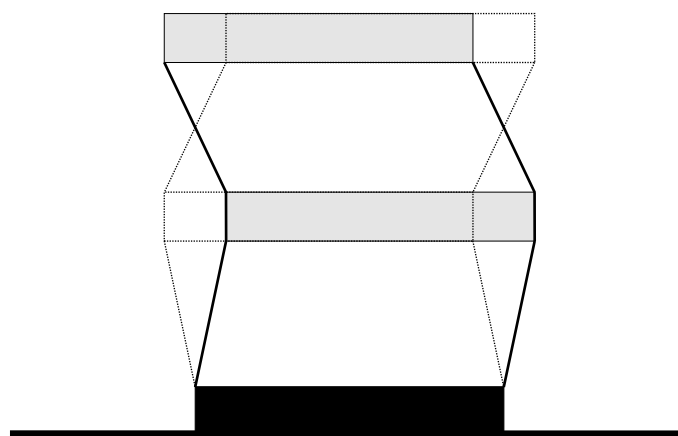


Figure 3 Second mode of vibration ($r_2=-1$).

The general solution of Equation (4) consists of the sum of the principal-mode solutions in Equations (14) - (17). Thus, the free vibration of two-DOF system is

$$x_1 = x_1(\omega_1) + x_1(\omega_2) = r_1 B_1 \sin(\omega_1 t + \phi_1) + r_2 B_2 \sin(\omega_2 t + \phi_2) \quad (18)$$

$$\text{and } x_2 = x_2(\omega_1) + x_2(\omega_2) = B_1 \sin(\omega_1 t + \phi_1) + B_2 \sin(\omega_2 t + \phi_2) \quad (19)$$

These expressions contain four arbitrary constants (B_1, B_2, ϕ_1, ϕ_2) and may be obtained by from the four given initial conditions of displacement and velocity of the two masses at time $t=0$, i.e.

$$\begin{array}{lll} x_1(0) = x_{10} & \dot{x}_1(0) = v_{10} & \text{for mass } m_1 \\ x_2(0) = x_{20} & \dot{x}_2(0) = v_{20} & \text{for mass } m_2 \end{array}$$

Equations (18) and (19) represent rather complicated motions that are not periodic unless the natural frequencies ω_1 and ω_2 happen to be commensurate (i.e. $a\omega_1 = b\omega_2$, and a, b are integers). The system performs a pure harmonic motion only if carefully started in one of its principal modes.

In summary: Two-DOF system has two natural frequencies and two principal modes. Free vibration of the system is a combination of the two principal modes.

3.3 FORCED VIBRATION

Vibration cannot continue indefinitely without external forcing agency present. One common cause of vibration is due to the unbalance of rotating parts. The centre of gravity of an unbalanced mass m_0 has a radial eccentricity e from the geometric axis of rotation, as shown in Figure 4.

When the mass is rotating at an angular speed Ω , the unbalanced force in the x direction is

$$P(t) = m_0 \Omega^2 e \cos(\Omega t) \quad (20)$$

The unbalanced force is not a constant but varies with the square of the rotating speed.

Substituting Equation (20) into Equation (2) gives

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} 2k_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m_0 \Omega^2 e \\ 0 \end{bmatrix} \cos \Omega t \quad (21)$$

The particular solutions of these equations are

$$x_1 = A \cos(\Omega t) \quad x_2 = B \cos(\Omega t) \quad (22)$$

where A and B are the amplitudes of steady-state responses of masses m_1 and m_2 .

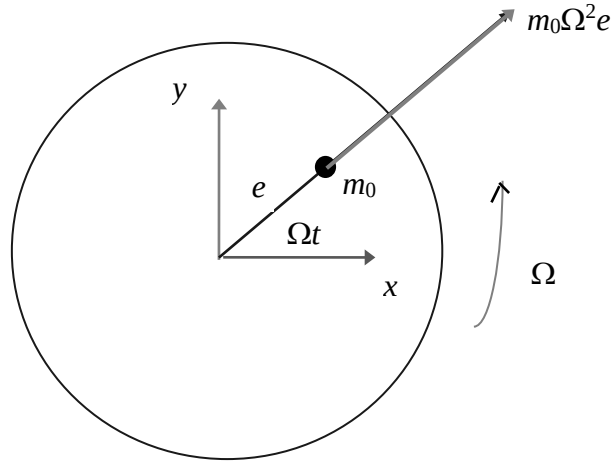


Figure 4 A shaking force of $m_0 \Omega^2 e$ is produced due to the unbalanced mass m_0 rotating with angular speed Ω .

Substituting Equations (22) into (21) produces the following algebraic equations,

$$\begin{bmatrix} 2k_1 - \Omega^2 m_1 & -2k_1 \\ -2k_1 & 2(k_1 + k_2) - \Omega^2 m_2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} m_0 \Omega^2 e \\ 0 \end{bmatrix} \quad (23)$$

The value of the determinant is the frequency equation for the system, which we solved earlier (see Equation (9)).

$$m_1 m_2 \omega^4 - [2m_1(k_1 + k_2) + 2m_2 k_1] \omega^2 + 4k_1 k_2 = m_1 m_2 (\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2) \quad (24)$$

The solutions for Equation (23) can be obtained

$$A = \frac{m_0 \Omega^2 e [2(k_1 + k_2) - \Omega^2 m_2]}{m_1 m_2 (\Omega^2 - \omega_1^2)(\Omega^2 - \omega_2^2)} \quad (25)$$

$$B = \frac{2m_0 \Omega^2 e k_1}{m_1 m_2 (\Omega^2 - \omega_1^2)(\Omega^2 - \omega_2^2)} \quad (26)$$

Substituting Equations (25) and (26) into (22) gives the vibration displacements for masses 1 and 2

$$x_1 = \frac{m_0 \Omega^2 e [2(k_1 + k_2) - \Omega^2 m_2]}{m_1 m_2 (\Omega^2 - \omega_1^2)(\Omega^2 - \omega_2^2)} \cos(\Omega t) \quad (27)$$

$$x_2 = \frac{2m_0 \Omega^2 e k_1}{m_1 m_2 (\Omega^2 - \omega_1^2)(\Omega^2 - \omega_2^2)} \cos(\Omega t) \quad (28)$$

Equations (27) and (28) show that masses m_1 and m_2 will vibrate at the same frequency as Ω , which is the rotating speed of unbalanced mass. The amplitudes become infinitely large when $\Omega = \omega_1$ or $\Omega = \omega_2$. Thus, for two-DOF system there are two conditions of resonance, one corresponding to each of the two natural frequencies of free vibration.

The amplitude ratio of two masses can be calculated from Equations (27) and (28) as

$$\frac{A}{B} = \frac{[2(k_1 + k_2) - \Omega^2 m_2]}{2k_1} \quad (29)$$

When the system vibrates at $\Omega = \omega_1$ (first resonance), this ratio approaches r_1 , given in Equation (12). Similarly, at the second resonance ($\Omega = \omega_2$), the ratio approaches r_2 , given in Equation (13). This result means that for each condition of resonance the forced vibration is in the corresponding principal mode.

In summary: Two-DOF system has two resonant conditions. At resonance, vibration of the two masses must follow the corresponding principal mode.

4. EQUIPMENT

4.1 Equipment List

The instruments used are listed below:

- Accelerometers
- Charge amplifier
- Storage Oscilloscope
- D. C. power supply
- Printer

4.2 Equipment technical data

The physical properties of the rig are

Mass: $m_1 = 1.40$ kg (including the motor), $m_2 = 1.06$ kg
 Stiffness: $k_1 = 8800$ N/m, $k_2 = 8520$ N/m

4.3 Experimental Setup

The experimental set-up is shown in Figure 5. The vibration rig comprises two masses connected at both ends by two steel rulers. A variable speed d.c. motor with an unbalanced rotating disc is mounted on mass 1. The rotation speed of the motor can be changed by adjusting the voltage to the motor. Two accelerometers are attached onto mass 1 and mass 2 to measure the vibration. The vibration signal is amplified by the charge amplifier and the results are displayed in the oscilloscope.

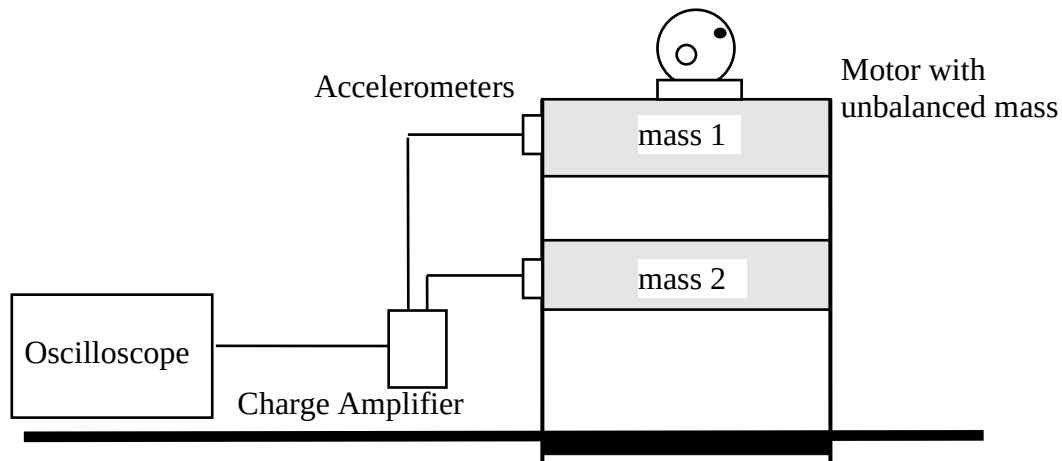


Figure 5 Experimental set-up

5. EXPERIMENT PROCEDURE

5.1 Experiment 1: Free Vibration of Two-DOF System

- Fix two accelerometers onto mass 1 and mass 2.
- Strike mass 1 horizontally by using a hammer, and observe the two vibration displacement signals on the oscilloscope.
- Print or take a photo of the vibration displacements against time waveforms.
- Strike mass 2 and repeat the above measurement from a) to c).
- Read Questions 1 to 3 in Section 6 and make sure you have sufficient information to answer them.

5.2 Experiment 2: Forced Vibration

- Fix the two accelerometers at mass 1 and mass 2.
- Vary the d.c.voltage of the motor from 3V to 14 V so that different frequencies can be obtained. Take the motor frequency from the oscilloscope trace. Take more readings in the region of the resonances. The first resonance should occur at about 3 V and the second should occur at about 10 V.
- Measure vibration amplitudes (peak-to-peak) of mass 1 and mass 2 and corresponding motor rotating speeds (Hz).
- Read Question 4 in Section 6 and make sure you have sufficient information to answer it.

5.3 Experiment 3: Vibration Modes

- a) Tune motor speed to the first resonant frequency.
- b) Observe the vibration mode. Do the two masses vibrate in phase or out of phase?
- c) Tune the motor speed to the second resonant frequency and Observe the vibration mode.
- d) Read Question 5 in Section 6 and make sure you have sufficient information to answer it.

6. RESULTS & DISCUSSION

Experiment 1: Free Vibration of Two-DOF System

Present the printed results for Experiment 1 and discuss Questions 1, 2 and 3.

Question 1: Does the system vibrate harmonically? Why not?

Question 2: Does striking on mass 1 and mass 2 produce similar vibration waveforms? Explain your answers.

Question 3: Can you use the vibration amplitude against time plots to measure the natural frequencies of the system? How do you measure it?

Experiment 2: Forced Vibration

Tabulate the measurement results for forced vibration. Plot a graph by using the vibration levels for mass 1 and mass 2 as y-axis and the corresponding motor speed as x-axis. Discuss Question 4.

Question 4: Suppose the test rig represents a two storey building. By installing a motor on the roof (mass 1), discuss the potential vibration problems and recommend actions to reduce the vibration level. (The discussion must refer to the measurement results).

Experiment 3: Vibration Modes

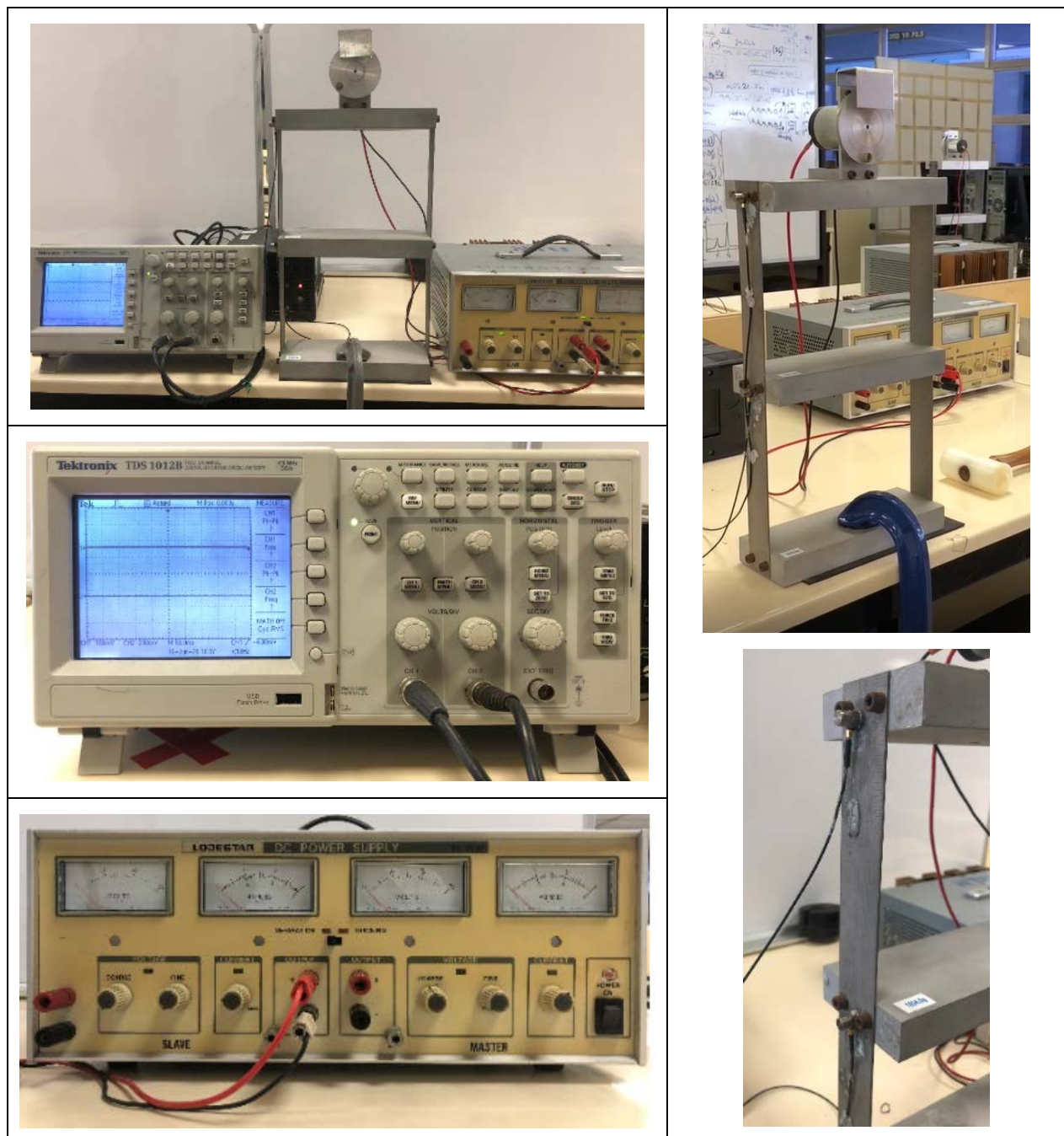
Question 5: Would vibration level increase, decrease or remain unchanged if the motor is installed at mass 2? Explain your answer for motor speed close to the resonances.

7. REFERENCES

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